Naming Convention in Java, Exceptions, Benefits of OOP

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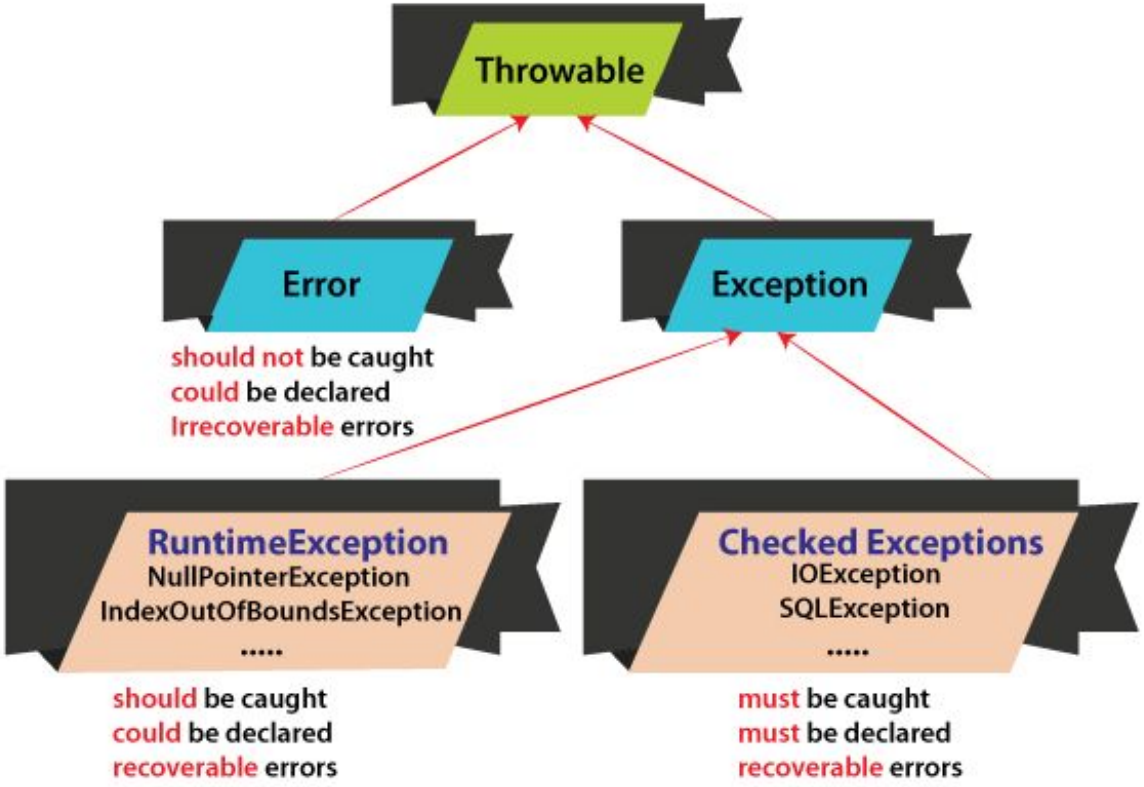
Identifiers Type	Naming Rules	Examples
Class	• It should start with the **uppercase** letter. • If the name contains multiple words, then the first letter of each internal word capitalized. • It should be a **noun** such as Color, Button, System, Thread, etc. • Use appropriate words, instead of **acronyms**.	**class** Raster; **class** MachineLearning;
Interface	• Interface names should be capitalized like class names. • It should be an **adjective** such as Runnable, Remote, ActionListener.	**interface** RasterDelegate; **interface** Storing;

Method	• It should start with lowercase letter.
	• It should be a **verb** such as main(), print(), println().
	• If the name contains multiple words, start it with a lowercase letter followed by an uppercase letter.
		run();
		runFast();
		getBackground();

Identifiers Type	Naming Rules	Examples
Variable	• It should start with a lowercase letter such as id, name. • It should not start with a digit. • It should not start with the special characters like & (ampersand), $ (dollar), _ (underscore). • Should not be a keyword. (Like **void**) • White space not allowed. (Like **var name**) • If the name contains multiple words, start it with the lowercase letter followed by an uppercase letter like-firstName, lastName. • Avoid using one-character variables such as x, y, z. • Can contain alphabet, special characters and digits if the other conditions are met.	int i; char c; float myWidth;
Package	• It should be a lowercase letter such as java, lang. • If the name contains multiple words, it should be separated by dots (.) such as java.util, java.lang.	package **com.java;**
Constant	• It should be in uppercase letters such as RED. • If the name contains multiple words, it should be separated by an underscore(_) such as MAX_PRIOR. • It may contain digits but not as the first letter.	static final int MIN_WIDTH = 4; static final int MAX_WIDTH = 999;

Basis of Comparison	Exception	Error
Recoverable/ Irrecoverable	Exception can be recovered by using the try-catch block.	An error cannot be recovered.
Type	It can be classified into two categories i.e. checked and unchecked.	All errors in Java are unchecked.
Occurrence	It occurs at compile time or run time.	It occurs at run time.
Package	It belongs to java.lang.Exception package.	It belongs to java.lang.Error package.
Known or unknown	Only checked exceptions are known to the compiler.	Errors will not be known to the compiler.

Basis of Comparison	Exception	Error
Causes	It mainly occurs in the code written by the developers.	It is mainly occur due to the lack of system resources.
Example	**Checked Exceptions:** SQLException, IOException **Unchecked Exceptions:** ArrayIndexOutOfBoundException, NullPointerException, ArithmaticException	**Unchecked Exceptions:** Java.lang.StackOverFlow, java.lang.OutOfMemoryError


Throwable
Error
Exception
should not be caught
could be declared
Irrecoverable errors
RuntimeException
NullPointerException
IndexOutOfBoundsException
.....
Checked Exceptions
IOException
SQLException
.....
should be caught
could be declared
recoverable errors
must be caught
must be declared
recoverable errors

Example of Error

```java
public class ErrorExample {
   public static void main(String[] args){
      recursiveMethod(10)
   }
   public static void recursiveMethod(int i){
      while(i!=0){
         i=i+1;
         recursiveMethod(i);
      }
   }
}
```

Output

```
Exception in thread "main" java.lang.StackOverflowError
   at ErrorExample.ErrorExample(Main.java:42)
```

Example of Exception

```java
public class ExceptionExample {
   public static void main(String[] args){
      int x = 100;
      int y = 0;
      int z = x / y;
   }
}
```

Output

```
java.lang.ArithmeticException: / by zero
   at ExceptionExample.main(ExceptionExample.java:7)
```

Exception and Error

- Error, and Exception both are subclasses of the Java Throwable class that belongs to java.lang package.
- When an error is detected, an exception is thrown.
- Any exception that is thrown must be caught by the **exception handler**.
- If the programmer has forgotten to provide an exception handler, the exception will be caught by the catch-all exception handler provided by the system.
- Exception may be rethrown if exception handler is failure to handle it.

Benefits of OOP

- **Reusability**: In OOP's programs, functions, and modules that are written by a user can be reused by other users without any modification.
- **Inheritance**: Through this we can eliminate redundant code and extend the use of existing classes.
- **Data Hiding**: The programmer can hide the data and functions in a class from other classes. It helps the programmer to build the secure programs.
- **Reduced complexity of a problem**: The given problem can be viewed as a collection of different objects. Each object is responsible for a specific task. The problem is solved by interfacing the objects. This technique reduces the complexity of the program design.

Benefits of OOP

- **Easy to Maintain and Upgrade**: OOP makes it easy to maintain and modify existing code as new objects can be created with small differences to existing ones. Software complexity can be easily managed.
- **Message Passing**: The technique of message communication between objects makes the interface with external systems easier.
- **Modifiability**: it is easy to make minor changes in the data representation or the procedures in an OO program. Changes inside a class do not affect any other part of a program, since the only public interface that the external world has to a class is through the use of **methods**.